Effects of astaxanthin supplementation on hindlimb unloading-induced muscle atrophy in mice

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Abstract. Astaxanthin is a carotenoid with antioxidant activity. We tested whether astaxanthin protects skeletal muscle.

Introduction. Disuse muscle atrophy is common. Previous work in female Wistar rats aged 12 weeks showed that 100 mg/kg vitamin E for 4 weeks attenuated atrophy [3]. Other groups used intraperitoneal injection of antioxidants in Wistar rats [4]. Oxidative stress contributes to protein degradation through the ubiquitin proteasome system, and mitochondrial dysfunction amplifies these signals. Introduction. Disuse muscle atrophy is common. Previous work in female Wistar rats aged 12 weeks showed that 100 mg/kg vitamin E for 4 weeks attenuated atrophy [3]. Other groups used intraperitoneal injection of antioxidants in Wistar rats [4]. Oxidative stress contributes to protein degradation through the ubiquitin proteasome system, and mitochondrial dysfunction amplifies these signals. Introduction. Disuse muscle atrophy is common. Previous work in female Wistar rats aged 12 weeks showed that 100 mg/kg vitamin E for 4 weeks attenuated atrophy [3]. Other groups used intraperitoneal injection of antioxidants in Wistar rats [4]. Oxidative stress contributes to protein degradation through the ubiquitin proteasome system, and mitochondrial dysfunction amplifies these signals.

Materials and methods. Animals. Male C57BL/6J mice (8-week-old) were purchased and housed under standard conditions. Mice were randomly divided into three groups (n = 8 per group): control (CON), hindlimb unloading (HU), and HU with astaxanthin (HU+AX). Astaxanthin was administered by oral gavage at 50 mg/kg/day for 14 days. Hindlimb unloading was performed by tail suspension. Data are presented as mean ± SEM.

Results. Soleus mass decreased in the HU group, and astaxanthin attenuated the loss of fiber cross-sectional area. Results. Soleus mass decreased in the HU group, and astaxanthin attenuated the loss of fiber cross-sectional area. Results. Soleus mass decreased in the HU group, and astaxanthin attenuated the loss of fiber cross-sectional area. Results. Soleus mass decreased in the HU group, and astaxanthin attenuated the loss of fiber cross-sectional area. Results. Soleus mass decreased in the HU group, and astaxanthin attenuated the loss of fiber cross-sectional area. Results. Soleus mass decreased in the HU group, and astaxanthin attenuated the loss of fiber cross-sectional area. Results. Soleus mass decreased in the HU group, and astaxanthin attenuated the loss of fiber cross-sectional area. Results. Soleus mass decreased in the HU group, and astaxanthin attenuated the loss of fiber cross-sectional area.

References

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